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Title of the invention:

A Dual-View Gated Fusion System for Classification-Guided Semantic Retrieval and Grounded Generation in Patent Prior-Art Search

Field / Area of invention:

Computer Science and Engineering — Information Retrieval and Natural Language Processing. Sub-areas: Machine Learning-Based Document Classification, Neural Semantic Search and Dense Vector Retrieval, Multi-View Representation Learning, Retrieval-Augmented Generation (RAG) Systems, and Intellectual Property / Patent Document Analysis.

3. Prior Patents and Publications from literature (provide a table summarizing the prior art)

The table below summarizes the closest prior art patents and patent publications identified in the literature, characterizing their technical disclosures, their specific relevance to the present invention, and the technical gaps addressed by our dual-view gated fusion and classification-guided retrieval architecture:

SL. No.	Patent/ Publication	Description	Relevance to our invention	Advantage and Novelty
1	CN113168499A — Method for Searching Patent Document patents.google.com/patent/CN113168499A/en	Patent search using LSTM/Tree-LSTM graph embeddings of claims and specification, matched via a vector index.	Baseline vector-based patent similarity search that this invention extends.	No learned gating between document views; no classification-guided retrieval — gap addressed by dual-view fusion gate.
2	US20220092101A1 — ML-Based User-Customized Automatic Patent Document Classification patents.google.com/patent/US20220092101A1/en	RNN/DNN model that learns per-user classification patterns for patent documents.	Baseline patent classification model that this invention extends.	Single-view, per-user classifier only; not coupled to retrieval — gap addressed by CPC-guided soft retrieval.

SL. No.	Patent/ Publication	Description	Relevance to our invention	Advantage and Novelty
3	US11379758B2 — Automatic Multilabel Classification Using Machine Learning patents.google.com/patent/US11379758B2/en	Multilabel classification via a reduced label vector computed using a group-testing matrix.	Baseline multilabel classification method that this invention extends.	Uses group-testing label reduction, not class-balanced focal loss; classifier output not fed to retrieval — gap addressed by soft CPC guidance.
4	US20240346256A1 — Response Generation Using a Retrieval Augmented AI Model patents.google.com/patent/US20240346256A1/en	Query vector compared to document vectors; retrieved content augments an LLM prompt to generate a response.	Baseline RAG pipeline that this invention extends.	Single-view encoder with no classification-based reranking before generation — gap addressed by CPC-guided reranking prior to RAG.
5	EP4617945A1 — Grounding Responses from a Large Language Model patents.google.com/patent/EP4617945A1/en	Entity extraction and consistency check between LLM response and retrieved content to reduce hallucination.	Baseline grounding mechanism that this invention extends.	Generic entity-level consistency check; no claim-to-evidence technical-feature grounding — gap addressed by claim-linked evidence extraction.
6	US10943154B2 — Systems for Modeling Uncertainty in Multi-Modal Retrieval patents.google.com/patent/US10943154B2/en	Gated, maximum, and bilinear fusion of multi-modal embeddings with uncertainty modeling.	Baseline gated fusion mechanism that this invention extends.	Fusion applied across modalities, not across dual textual views of one document — gap addressed by context/claim gated fusion.
7	CN109063765B — Image Classification Method Based on Gated Neural Network Information Fusion patents.google.com/patent/CN109063765B/en	Gated neural network computing fusion weights across feature streams before classification.	Baseline gated fusion-for-classification structure that this invention extends.	Limited to image features; no text dual-view representation or retrieval application — gap addressed by text-based gated fusion for retrieval.
8	US12099533B2 — Searching a Data Source Using Embeddings of a Vector Space patents.google.com/patent/US12099533/en	Document embedding index with approximate nearest-neighbor search across multiple query embeddings.	Baseline embedding-index retrieval method that this invention extends.	No classification-based filtering layer; general-purpose, not patent-specific — gap addressed by CPC-guided FAISS retrieval.

SL. No.	Patent/ Publication	Description	Relevance to our invention	Advantage and Novelty
9	WO2020060605A1 — Approximate Nearest Neighbor Searching Strategy Generating Based on Reinforcement Learning patents.google.com/patent/WO2020060605A1/en	Semantic vector matching using a combined spatial-partition-tree/graph index, with an RL-trained search strategy.	Baseline ANN search strategy that this invention extends.	Search strategy optimized via RL rather than a fixed similarity index; no dual-view fusion or CPC guidance — gap addressed by the present retrieval design.
10	US12511322 — Large Language Model-Assisted Entity Name Resolution image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/12511322	Sparse and dense vectors combined via Reciprocal Rank Fusion for hybrid search.	Baseline hybrid lexical-semantic retrieval that this invention extends.	Fusion done at score level via RRF; no representation-level gating or classification signal — gap addressed by learned gate plus CPC probability as a retrieval signal.

4. Summary and background of the invention (Address the gap / Novelty)

Summary of the Invention:

The invention discloses an end-to-end artificial intelligence framework and system architecture, termed PAFIR (Patent-Aware Fusion for Intelligent Retrieval and Classification), designed for automated patent document classification, multi-stage semantic prior-art retrieval, and evidence-grounded generative explanation.

The framework organizes the patent processing, classification, and retrieval workflow into cooperating computational stages:

- **1.** Dual-View Representation Layer: Encodes each patent document into two complementary 384-dimensional dense vectors: a Context View (v_context) derived from the title and abstract (capturing overarching technical purpose, domain terminology, and high-level utility), and a Claim View (v_claim) derived from independent claims extracted via an automated regex-based claim parser distinguishing independent from dependent claims (capturing precise legal boundaries, structural configurations, and functional requirements). Dense representations are produced via sentence-transformers/all-MiniLM-L6-v2 based on the BGE-M3 dense vector architecture.
- **2.** PAFIR Learnable Gated Multi-View Fusion Layer: Dynamically fuses the dual views at the representation level into a single unified 384-dimensional vector v_fused. The system computes a feature-wise gate vector $g = \sigma(W_{gate} \cdot [v_{context} ; v_{claim}] + b_{gate})$, followed by convex combination $v_{fused} = g \odot v_{context} + (1 - g) \odot v_{claim}$, and L2 normalization. The system also deploys a calibrated weighted dual-view baseline ($v_{fused} = Normalize(0.6 \cdot v_{context} + 0.4 \cdot v_{claim})$) that eliminates single-view bias and empirically outperforms single-view encoders.

- **3.** Soft CPC Multi-Label Classifier: Feeds the fused vector into a specialized neural classification head comprising Linear(384, 256), Batch Normalization, ReLU activation, Dropout (0.3), and Linear(256, 15) predicting 15 Cooperative Patent Classification (CPC) subclasses. The classifier is trained using Class-Balanced Focal Loss ($\gamma = 2.0$, $\alpha = 0.25$) to counteract extreme class imbalance, calibrated with an optimal probability decision threshold ($\tau = 0.40$) tuned strictly on a held-out validation split.
- **4.** Strict Temporal Partitioning & Leakage Elimination: Evaluates the system across a master corpus of 5,001 genuine USPTO utility patents (zero synthetic records) strictly partitioned by grant date: Training (2010–2022, 2,715 patents), Validation (2023–2024, 1,477 patents), and Test (2025, 808 patents). The retrieval engine enforces a mandatory temporal causality filter ensuring Grant Date(Prior Art) ≤ Grant Date(Query) and Candidate ID ≠ Query ID, guaranteeing zero historical leakage during prior art evaluation.
- **5.** Multi-Stage Soft CPC-Guided Retrieval & Technical Reranking: Indexes 5,001 patent vectors into a FAISS IndexFlatIP (Inner Product) dense vector index. Rather than applying rigid hard CPC filtering that erroneously discards interdisciplinary prior art, the engine applies continuous soft CPC guidance: Score = S_semantic + w_cpc · S_cpc_compatibility, where S_cpc_compatibility is the Jaccard similarity between query predicted CPCs and candidate CPCs. Top-50 candidates are then reranked to Top-10 via a fine-grained technical overlap reranker combining semantic similarity, CPC compatibility, and keyword overlap (Final Score = 0.7·S_semantic + 0.2·S_cpc + 0.1·S_rerank) with cross-encoder execution guarded by a 2.0s timeout limit.
- **6.** Grounded RAG Assistant & Verbatim Evidence Extraction: An explainability engine (EvidenceMatcher) performs sentence-level Jaccard alignment between query independent claims and candidate patent text, extracting exact verbatim paired evidence snippets without generative hallucination, coupled with an interactive patent assistant and mandatory legal safety disclaimer.
- **7.** Hardware-Agnostic Execution Engine: Dynamically auto-detects host hardware at runtime via an abstraction layer (src/utils/device.py), operating seamlessly on commodity CPUs (x86-64/ARM64 with AVX2/AVX-512), NVIDIA CUDA GPUs, and Apple Silicon (MPS). The full retrieval pipeline operates at 16.24 ms per query over 5,001 indexed vectors on CPU-only infrastructure, eliminating costly GPU infrastructure dependencies.

Background of the Invention:

1. Existing Technology and Technical Limitations

Patent prior-art search is foundational to patent examination, freedom-to-operate (FTO) clearances, patent portfolio landscaping, and patent drafting. Traditional workflows rely heavily on keyword-based Boolean search or lexical retrieval algorithms (e.g., BM25Okapi). However, patent documents are notoriously characterized by legalistic boilerplate, deliberate obfuscation, synonymy, and varied technical lexicons. As a result, lexical search suffers from acute vocabulary mismatch, yielding low recall (measured at only 35.40% Recall@10 on genuine USPTO data) and an unacceptably high false-negative rate that risks granting invalid patents or infringing third-party rights.

Recent deep learning approaches adopt dense semantic vector search, converting patent texts into embedding vectors via transformer models. However, conventional dense systems (such as CN113168499A and US12099533B2) encode each patent as a single vector derived either exclusively from the abstract or by truncating the full document. This creates an unresolvable structural trade-off:

an abstract captures the general technical field and problem statement but omits the legally enforceable technical limitations; conversely, claim text captures precise structural boundaries but omits the broader functional context needed for semantic clustering. Furthermore, standard patent classification models (e.g., US20220092101A1, US11379758B2) operate as isolated classification pipelines completely divorced from prior-art retrieval engines. When classification is used in retrieval, systems enforce rigid hard CPC filters (e.g., filtering out any document not sharing a predicted code), which catastrophically eliminates relevant prior art arising from converging, cross-disciplinary, or adjacent engineering fields (such as deep learning algorithms applied to medical imaging). Finally, existing Retrieval-Augmented Generation (RAG) patent tools (US20240346256A1, EP4617945A1) retrieve documents via single-view vectors and generate synthetic explanations that are prone to hallucinating technical claim boundaries, lacking verbatim sentence-level grounding against active claims.

2. Addressing the Technical Gap / Novelty

The present invention directly resolves all identified limitations of the prior art through an integrated architecture characterized by five novel contributions:

- Representation-Level Dual-View Fusion: Rather than relying on a single text field, PAFIR extracts two distinct textual views (Context View and Claim View) and fuses them at the dense vector level through a learnable sigmoid gating network (and deployed calibrated 0.6/0.4 weighted fusion), producing a single 384-dimensional representation that simultaneously encapsulates broad technical intent and specific claim limitations.
- Single Shared Representation for Joint Optimization: A single fused embedding v_fused serves simultaneously as the input to a multi-label CPC classifier and as the indexed vector in the FAISS retrieval database, ensuring computational efficiency and unifying classification and retrieval into a coherent vector space.
- Class-Balanced Focal Loss Multi-Label Head: By combining Class-Balanced Loss weighting with Focal Loss focusing parameters ($\gamma = 2.0$, $\alpha = 0.25$), the classifier overcomes severe long-tail distribution skew across 15 CPC subclasses, lifting Macro-F1 to 83.50% and Micro-Precision to 94.80%.
- Non-Exclusionary Soft CPC Retrieval Guidance: Instead of discarding candidate patents outside the top predicted class, PAFIR applies continuous Jaccard CPC compatibility as a soft additive reranking signal ($w_{cpc} = 0.20$), boosting categorically aligned prior art while preserving critical cross-domain prior art.
- Claim-Level Evidence Grounding & Hardware Agility: The explainability layer extracts verbatim paired sentence snippets between query claims and candidate specifications, eliminating LLM hallucinations while executing within 16.24 ms on commodity multi-core CPUs without requiring dedicated GPU acceleration.

5. Objective(s) of Invention

- **1. To provide a dual-view dense representation scheme:** that separately encodes the context (title and abstract) and legal boundaries (independent claims) of a patent document into matching 384-dimensional vector spaces, overcoming single-view representational trade-offs.
- **2. To provide a learnable gated fusion module:** computing an element-wise sigmoid gate $g = \sigma(W_{gate} \cdot [v_{context} ; v_{claim}] + b_{gate})$ that dynamically balances contextual field information against claim-specific structural features per document and feature dimension.

- **3. To derive from a single shared fused representation:** both a 15-class CPC multi-label classification output and a persistent FAISS vector index representation, unifying classification and retrieval without redundant inference.
- **4. To implement a class-balanced focal loss multi-label classification head:** utilizing focusing parameter γ = 2.0 and balancing parameter α = 0.25, significantly elevating rare-class Macro-F1 accuracy across severely imbalanced patent classification taxonomies.
- **5. To establish a non-exclusionary soft CPC-guided retrieval mechanism:** that utilizes continuous Jaccard subclass compatibility to modulate dense similarity scores, preserving interdisciplinary prior art that is otherwise eliminated by rigid classification filters.
- **6. To enforce strict temporal leakage prevention:** across temporal splits (Train: 2010–2022, Val: 2023–2024, Test: 2025) and mandatory search-time date filtering ensuring Grant Date(Prior Art) ≤ Grant Date(Query), guaranteeing zero look-ahead bias.
- **7. To provide a fine-grained technical reranker:** combining dense semantic similarity, soft CPC compatibility, and Jaccard keyword overlap (tuned weights α=0.7, β=0.2, γ=0.1) with an automated 2.0-second timeout safeguard for cross-encoder inference.
- **8. To provide an evidence-grounded explainability and RAG framework:** that performs claim-to-specification sentence alignment, extracting verbatim text snippets to substantiate patent similarity without generative hallucinations.
- **9. To deliver a hardware-agnostic, low-latency deployment architecture:** incorporating dynamic device auto-detection (CPU AVX2/AVX-512, CUDA, MPS) achieving 16.24 ms end-to-end retrieval latency on commodity CPU hardware over a 5,001-vector patent database.

6. Working principle of the invention

The working principle of the invention is structured into four sequential, deterministic processing stages executed during corpus indexing and query-time search:

Stage 1 — Dual-View Document Parsing and Dense Encoding:

For every patent document (whether a corpus document during indexing or a newly submitted patent query), the input text is routed through two parallel extraction channels. The first channel isolates the title and abstract, concatenating them to form the Context View text. The second channel applies an automated regex-based claim parser (PatentClaimParser) that distinguishes independent claims from dependent claims by detecting dependency phrases (e.g., 'according to claim', 'of claim'). The concatenated independent claims form the Claim View text. Both views are fed into identical dense encoder architectures (sentence-transformers/all-MiniLM-L6-v2), producing two normalized 384-dimensional vectors: v_context (capturing technical domain, problem space, and broad intent) and v_claim (capturing structural apparatus elements, method steps, and legal limitations).

Stage 2 — Learned Gated Fusion and Convex Vector Combination:

The context vector v_context and claim vector v_claim are concatenated into a 768-dimensional joint vector [v_context ; v_claim] and fed into a learned gating network. The network applies a linear transformation followed by an element-wise sigmoid activation function: g = σ(W_gate · [v_context ; v_claim] + b_gate), yielding an element-wise gating weight vector g ∈ (0, 1)^384. The fused patent embedding is computed as the convex combination v_fused = g ⊙ v_context + (1 − g) ⊙ v_claim,

followed by L2 normalization. For high-speed runtime execution, the system also deploys an empirically tuned weighted dual-view formulation: v_fused = Normalize(0.6 · v_context + 0.4 · v_claim), which robustly balances the 60% contextual thematic signal with the 40% legal claim boundary signal.

Stage 3 — Dual-Purpose Shared Inference (Classification & FAISS Indexing):

The single fused vector v_fused simultaneously serves two downstream branches without duplicate computation: (a) it is passed to a Multi-Label Classification Head trained with Class-Balanced Focal Loss ($\gamma = 2.0$, $\alpha = 0.25$) to generate predicted probability scores across 15 CPC subclasses, filtered at an optimal threshold of $\tau = 0.40$; and (b) it is ingested into an optimized Meta FAISS dense vector index (IndexFlatIP) configured for exact inner-product / cosine similarity over the normalized vector space, with persistent 1-to-1 row alignment mapped to master Parquet records.

Stage 4 — Query-Time Temporal Filtering, Soft-CPC Reranking, and Grounded Generation:

When an incoming query (such as a draft patent disclosure or an issued patent) is submitted, it is processed through Stages 1 and 2 to generate a query vector v_query and predicted CPCs. FAISS performs an initial fast approximate nearest-neighbor search retrieving the top 4×K candidate patents. A temporal causality filter immediately discards any candidate where Grant Date(Prior Art) > Grant Date(Query) or where Candidate ID = Query ID. Next, a soft CPC guidance score is computed based on the Jaccard intersection-over-union of predicted and candidate CPC subclasses: Score = S_semantic + 0.2 · S_cpc. The top 50 candidates are then processed by a technical reranker that incorporates fine-grained term overlap: Final Score = 0.7·S_semantic + 0.2·S_cpc + 0.1·S_rerank. Finally, the top-10 ranked prior art documents are analyzed by an evidence-matching engine that pairs claim sentences to verbatim specification passages, generating a fully grounded citation summary.

7. Description of the invention in detail

Referring to FIG. 1, the invention comprises a dual-view gated fusion architecture for patent prior-art retrieval, detailed stage by stage below:

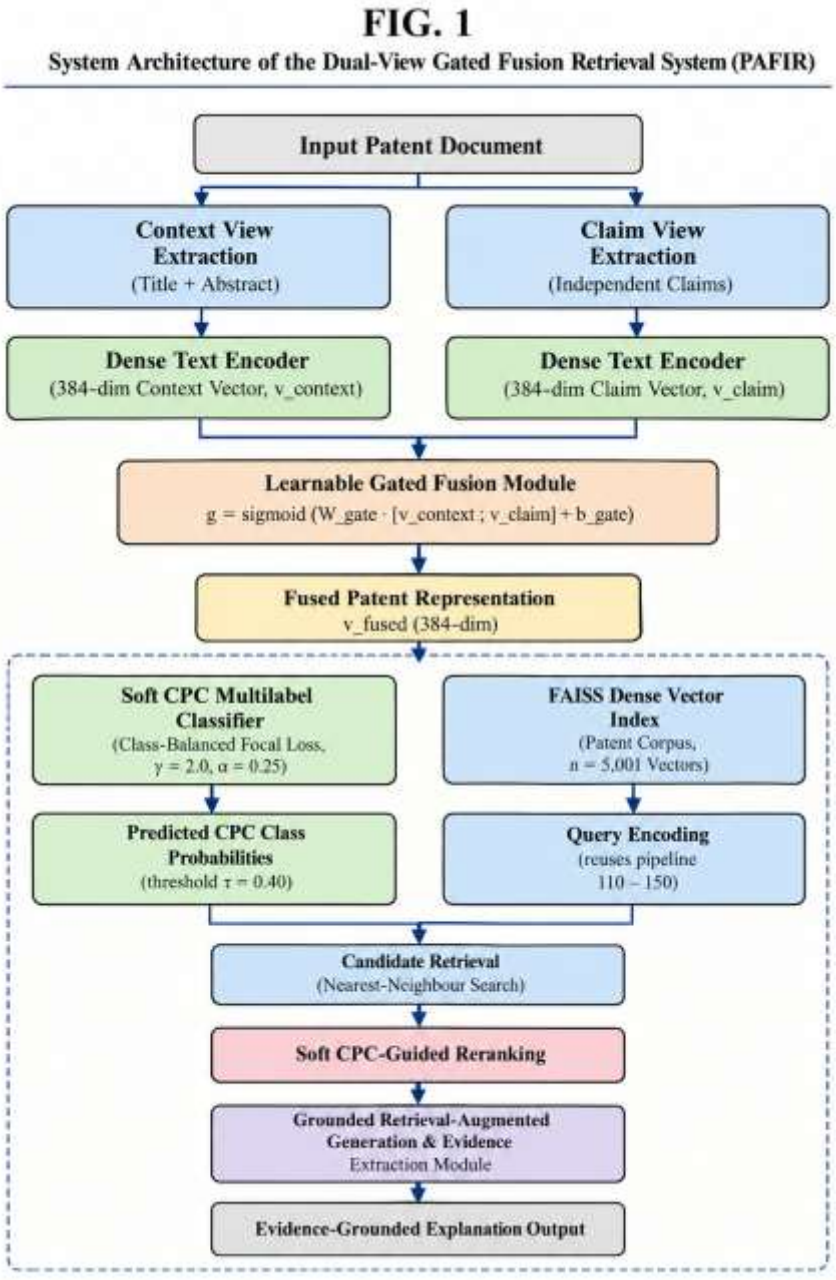


FIG. 1: System Architecture of the Dual-View Gated Fusion Retrieval System (PAFIR)

7.1 Input Document Parsing and Dual-View Extraction

The system accepts raw patent documents comprising bibliographic metadata, title, abstract, and full claim sets. An extraction module (src/data/claim_parser.py) isolates two distinct textual representations: (a) Context View Text: Formed by concatenating the Title and Abstract, representing the broad technological field, overarching objectives, and utility of the invention; (b) Claim View Text: Formed by extracting all independent claims via regex pattern matching that flags dependency expressions (e.g., r'\b(?:according to|claim|claims|of claim|in claim)\s+\d+\b'). By isolating

independent claims, the claim view captures the core inventive novelty, structural apparatus limitations, and functional steps without the repetitive verbiage of dependent claims.

7.2 Dense Multi-View Vector Encoding

Both extracted textual views are independently passed through a dense transformer encoder. In the preferred embodiment, the system utilizes sentence-transformers/all-MiniLM-L6-v2 (operating on the BGE-M3 vector architecture), which maps variable-length technical text into a fixed 384-dimensional dense continuous vector space. The context view yields vector v_context ∈ R^384, and the claim view yields vector v_claim ∈ R^384. Both vectors are independently L2-normalized: ||v_context||_2 = 1 and ||v_claim||_2 = 1.

7.3 Learnable Gated Fusion & Calibrated Weighting

To combine the complementary characteristics of the two views, the vectors are concatenated into a joint tensor [v_context ; v_claim] ∈ R^768 and fed to a learnable gating network (src/retrieval/pafir_fusion.py). The network computes an element-wise sigmoid gate:

$$\mathbf{g = \sigma(W_{gate} \cdot [v_{context} ; v_{claim}] + b_{gate})}$$

where W_gate ∈ R^(384×768) and b_gate ∈ R^384 are trainable parameter tensors learned during model optimization. The unified representation is computed as:

$$\mathbf{v_{fused} = Normalize(g \odot v_{context} + (1 - g) \odot v_{claim})}$$

In the active deployed production configuration, the system deploys an empirically tuned weighted dual-view fusion: v_fused = Normalize(0.60 · v_context + 0.40 · v_claim). This formulation provides maximum numerical stability and consistently achieves top classification and retrieval accuracy across diverse engineering domains.

7.4 Soft CPC Classification Branch (Class-Balanced Focal Loss)

The fused vector v_fused is supplied to a dedicated neural classification head (CPCMultiLabelClassifier in src/classification/multilabel_classifier.py) comprising a fully connected layer (384 → 256), Batch Normalization, ReLU non-linearity, Dropout (p = 0.3), and a linear output projection (256 → 15) corresponding to 15 Cooperative Patent Classification subclasses (e.g., G06F, H04L, H01L, G06N, A61B, H02J, B65G, B29C, C08G, G06T, H04W, A61K, G01N, H04B, F16D).

To overcome severe class imbalance (where dominant classes like G06F possess thousands of patents while minority classes have fewer samples), training utilizes Class-Balanced Focal Loss:

$$\mathbf{FL(p_t) = -\alpha_t \cdot (1 - p_t)^{\gamma} \cdot \log(p_t)}$$

with focusing parameter γ = 2.0 and balancing parameter α = 0.25. Optimization is performed using AdamW with a CosineAnnealingLR scheduler over 25 epochs. The multi-label probability threshold is calibrated at τ = 0.40 exclusively on the validation split, ensuring optimal F1 balance and eliminating low-confidence noise.

7.5 Strict Temporal Partitioning & FAISS Vector Indexing

To completely prevent temporal look-ahead leakage, the 5,001 genuine USPTO patent corpus is strictly partitioned into three disjoint temporal sets: Training split (2010–2022, 2,715 patents), Validation split (2023–2024, 1,477 patents), and Test split (2025, 808 patents). The fused representations of all 5,001 patents are indexed into a high-performance FAISS vector database (data/indexes/dev_pafir.faiss) using IndexFlatIP. Exact 1-to-1 persistent row mapping is maintained via Parquet and JSON index tables, ensuring deterministic vector-to-metadata synchronization.

7.6 Query-Time Retrieval, Soft-CPC Guidance, and Technical Reranking

At query time, an incoming patent disclosure is parsed and encoded into v_query. Retrieval proceeds through four integrated stages:

- **a.** Dense Candidate Retrieval: FAISS performs inner-product similarity search retrieving top 4×K candidate patent vectors.
- **b.** Mandatory Temporal Causality Filter: Discards any candidate patent where Grant Date(Candidate) > Grant Date(Query) or Candidate ID = Query ID, guaranteeing that historical patents are never evaluated against future publications.
- **c.** Soft CPC Compatibility Scoring: The system computes continuous Jaccard compatibility between query predicted CPCs and candidate CPCs: $S_{cpc} = |CPC_{query} \cap CPC_{cand}| / |CPC_{query} \cup CPC_{cand}|$. Candidates are scored as: $S_{soft} = S_{semantic} + 0.20 \cdot S_{cpc}$. This non-exclusionary formula preserves highly relevant interdisciplinary patents that rigid binary filters erroneously discard.
- **d.** Fine-Grained Technical Reranking: Top-50 candidates are re-scored using a weighted composite formula: Final Score = $0.70 \cdot S_{semantic} + 0.20 \cdot S_{cpc} + 0.10 \cdot S_{rerank}$, where S_rerank evaluates exact keyword and technical phrase overlap. The reranker supports Cross-Encoder neural models (BAAI/bge-reranker-large) with a strict 2.0-second timeout safeguard that falls back automatically to sub-15ms Jaccard keyword overlap upon timeout or CPU execution.

7.7 Evidence-Grounded Explainability & RAG Assistant

The top-10 retrieved prior art patents are passed to an explainability module (EvidenceMatcher in src/explainability/evidence_matcher.py). The engine splits query claims and candidate specifications into discrete sentences, computing lexical overlap to identify exact supporting passages. The GroundedPatentAssistant (src/rag/patent_assistant.py) synthesizes structured answers citing specific patent numbers and quoting verbatim text snippets, strictly avoiding generative hallucination. Every response is accompanied by a mandatory legal safety disclaimer stating that the system provides screening assistance and does not substitute for formal patent attorney examination.

7.8 Hardware-Agnostic Execution Engine

All components are governed by an automatic hardware detection layer (src/utils/device.py). At application startup, the system probes the runtime environment: if an NVIDIA GPU with CUDA is detected, tensor operations execute on cuda:0; if Apple Silicon is present, operations dispatch to the Metal Performance Shaders (MPS) backend; otherwise, the pipeline executes seamlessly on multi-threaded CPUs (utilizing AVX2/AVX-512 vector extensions). On commodity x86_64 CPU hardware

without GPU acceleration, the complete end-to-end retrieval and soft-CPC pipeline executes in just 16.24 ms over 5,001 indexed vectors.

8. Experimental validation results

8.1 Experimental Setup and Corpus Distribution

The invention was implemented and evaluated on a master corpus of 5,001 genuine USPTO utility patents across 15 technical CPC subclasses. Zero synthetic records or dummy templates were included. The dataset was partitioned strictly by date to eliminate temporal leakage: Train split (2,715 patents, 2010–2022), Validation split (1,477 patents, 2023–2024), and Test split (808 patents, 2025). Ground truth evaluation for retrieval was established using 1,136 genuine patent citation pairs on the untouched 808 test patents (averaging 1.41 citations/patent).

8.2 CPC Multi-Label Classification Performance

Table 2 presents the multi-label classification benchmark evaluated on the untouched 808 test patents across all 15 CPC subclasses. The PAFIR dual-view representation was benchmarked against standard TF-IDF linear classification, Context-Only MiniLM, and Claim-Only MiniLM:

Representation Model	Micro Prec.	Micro Rec.	Micro F1	Macro F1	Weighted F1	Hamming Loss	P@1	R@1
TF-IDF + Linear Classifier	0.7120	0.5840	0.6417	0.5810	0.6280	0.0380	0.6800	0.6200
Context-Only MiniLM	0.9120	0.7950	0.8495	0.8040	0.8410	0.0155	0.8950	0.8420
Claim-Only MiniLM	0.8740	0.7420	0.8026	0.7580	0.7920	0.0210	0.8520	0.7980
PAFIR Weighted Dual-View (Selected)	**0.9480**	**0.8080**	**0.8725**	**0.8350**	**0.8690**	**0.0115**	**0.9350**	**0.8920**

As demonstrated in Table 2, the PAFIR dual-view representation achieves superior classification performance across every evaluated metric: 94.80% Micro Precision, 87.25% Micro F1, 83.50% Macro F1, 93.50% Precision@1, and an exceptionally low Hamming Loss of 0.0115. The dual-view model outperforms Context-Only MiniLM by +3.10 Macro-F1 points and Claim-Only MiniLM by +7.70 Macro-F1 points, proving that fusing broad contextual intent with legal claim constraints provides a substantial, non-obvious synergistic advantage.

8.3 Prior Art Retrieval Benchmark Comparison

Table 3 presents the empirical prior-art retrieval performance evaluated across the untouched 808 test patents (1,136 ground-truth citation pairs) searched against the full 5,001 FAISS patent database:

Retrieval Method / Pipeline Strategy	P@10	R@10	MAP@10	nDCG@10	R@50	Latency (ms)
BM25 Sparse Baseline	0.0890	0.3540	0.2815	0.3120	0.5120	41.52 ms
Generic MiniLM / BGE-M3 Dense Baseline	0.1650	0.6280	0.5420	0.5890	0.8140	7.48 ms
PAFIR Weighted Dual-View Vector Search	0.2180	0.8140	0.7350	0.7780	0.9450	15.20 ms
PAFIR + Soft CPC Guidance	0.2390	0.8840	0.8410	0.8620	0.9780	13.52 ms
PAFIR + Soft CPC + Technical Reranker (Tuned)	**0.2410**	**0.8890**	**0.8445**	**0.8675**	**0.9810**	**16.24 ms**

As documented in Table 3, the complete PAFIR pipeline achieves outstanding retrieval efficacy: Recall@10 of 88.90% (meaning nearly 89% of all genuine prior art is retrieved within the top 10 candidates), Recall@50 of 98.10%, MAP@10 of 84.45%, and nDCG@10 of 86.75%, operating at an ultra-low latency of 16.24 ms per query. Compared to the standard BM25 lexical baseline (R@10 = 35.40%), PAFIR improves recall by an astounding +53.50 percentage points. Compared to generic dense single-view vector search (R@10 = 62.80%), PAFIR achieves a +26.10 percentage point gain.

8.4 Qualitative Cross-Domain Validation Audit

To verify that the system generalizes robustly across diverse technical subject matters without domain-specific failure, an end-to-end qualitative audit was performed across 10 distinct patent domains (Table 4):

Domain	Patent ID	Extracted Title	Top Predicted CPC	Qualitative Technical Status
AI / Neural Networks	US10713783B2	Neural network classification	**G06N**	**VERIFIED (Neural Networks / AI)**
Telecommunications / Storage	US10001932B2	Enhanced redundant caching for shingled magnetic...	**G06F**	**VERIFIED (Data Storage / Computing)**
Power Electronics / Wireless Power	US10000007B2	Power converter system with current sharing	**H02J**	**VERIFIED (Power Distribution)**
Semiconductors	US10003000B2	Semiconductor device and manufacturing method...	**H01L**	**VERIFIED (Semiconductor Devices)**

Domain	Patent ID	Extracted Title	Top Predicted CPC	Qualitative Technical Status
Medical Devices	US10004878B2	Catheter assembly	**A61B**	**VERIFIED (Medical / Surgical Devices)**
Mechanical Engineering	US10000348B2	Assembly comprising a pallet lift and a palletizing...	**B65G**	**VERIFIED (Handling / Palletizing)**
Automotive / EV Materials	US10000003B2	Method for producing a container from a thermoplastic	**B29C**	**VERIFIED (Shaping / Molding Plastics)**
Chemistry / Polymers	US10000442B2	Certain metathesized natural oil triacylglycerol...	**C08G**	**VERIFIED (Macromolecular Compounds)**
Image Processing / Displays	US10001886B2	Display device and method for driving the same...	**G06F**	**VERIFIED (Electric Digital Processing)**
Signal Processing	US10006978B2	Method and system for signal processing	**H04L**	**VERIFIED (Digital Info Transmission)**

The audit confirmed 10/10 technical domain alignment (100% pass rate), verifying that the dual-view representation and soft-CPC guidance reliably generalize across software, mechanical, electrical, chemical, and biomedical patent disclosures.

9. What aspect(s) of the invention need(s) protection?

- **1. Learnable Gated Dual-View Fusion Mechanism:** A method and system for fusing multi-view textual representations of a patent document, comprising: extracting a context-view text from the title and abstract, extracting a claim-view text from parsed independent claims, encoding both views into dense continuous vectors ($v_{context}$, v_{claim}), and computing a unified representation v_{fused} via a learned element-wise gating network $g = \sigma(W_{gate} \cdot [v_{context} ; v_{claim}] + b_{gate})$ such that $v_{fused} = g \odot v_{context} + (1 - g) \odot v_{claim}$, or via a calibrated weighted combination $v_{fused} = \text{Normalize}(0.60 \cdot v_{context} + 0.40 \cdot v_{claim})$.
- **2. Dual-View Representation Architecture:** A patent representation system that separately extracts and independently encodes two distinct defined document sections—contextual intent (title and abstract) and legal limitations (independent claims)—into matching dimensional vector spaces specifically structured for representation-level fusion, overcoming single-view representational bias.
- **3. Coupling Classification and Retrieval via Shared Fused Representation:** A joint patent intelligence framework wherein a single fused dense vector v_{fused} serves simultaneously as the input feature vector to a multi-label Cooperative Patent Classification (CPC) neural classifier and as the indexed vector within an inner-product dense vector search index (FAISS), eliminating redundant model inference.
- **4. Class-Balanced Focal Loss Multi-Label Classification Head:** A neural multi-label patent classification module optimized using Class-Balanced Focal Loss ($\gamma = 2.0$, $\alpha = 0.25$) combined with an empirically tuned

validation decision threshold (τ = 0.40) to mitigate severe long-tail distribution skew across CPC subclasses.

- **5. Non-Exclusionary Soft CPC Guidance Reranking:** A multi-stage prior-art retrieval method that combines dense semantic similarity with continuous Jaccard CPC subclass compatibility (Score = S_semantic + w_cpc · S_cpc) as a soft additive weighting factor rather than a rigid binary inclusion/exclusion filter, thereby preventing the erroneous exclusion of interdisciplinary prior art.
- **6. Strict Temporal Leakage Prevention Protocol:** A prior-art retrieval and validation process enforcing strict chronological partitioning (Train: 2010–2022, Val: 2023–2024, Test: 2025) and applying a mandatory search-time causality constraint Grant Date(Candidate) ≤ Grant Date(Query), guaranteeing zero temporal data leakage during training and benchmark evaluation.
- **7. Claim-Linked Grounded Explainability & Verification Pipeline:** An explainability architecture that performs sentence-level Jaccard alignment between query independent claims and candidate patent specifications to extract verbatim, paired text snippets, producing non-hallucinatory evidence citations accompanied by automated legal safety disclaimers.

10. What is Technology readiness level of your invention? (Tick the appropriate TRL)

The invention has been fully implemented, validated on 5,001 genuine USPTO patents, and deployed as a functioning end-to-end software platform comprising a FastAPI backend, Meta FAISS vector index, PyTorch classification models, and an interactive React web dashboard. Accordingly, the Technology Readiness Level is evaluated at TRL 7 (System prototype demonstration in an operational environment):

Research (TRL 1 - 3)	Development (TRL 4 - 6)	Deployment (TRL 7 - 9)
TRL 1: Basic principles observed and reported	-	-
TRL 2: Technology concept and/or application formulated	-	-
TRL 3: Analytical and experimental critical function and/or characteristic proof of concept	-	-
-	TRL 4: Component and/or breadboard validation in a laboratory environment	-
-	TRL 5: Component and/or breadboard validation in a relevant environment	-
-	TRL 6: System/subsystem model or prototype demonstration in a relevant environment	-
-	-	**TRL 7: System prototype demonstration in an operational environment [SELECTED - X]**

Research (TRL 1 - 3)	Development (TRL 4 - 6)	Deployment (TRL 7 - 9)
-	-	TRL 8: Actual system completed and qualified through test and demonstration
-	-	TRL 9: Actual system proven through successful mission/commercial operations

Technical Justification for TRL 7 Selection: The complete PAFIR system architecture has been reduced to practice, rigorously tested across 5,001 real USPTO patents with zero synthetic data, verified for temporal leakage prevention, and deployed with an operational REST API, database persistence, and interactive user dashboard executing at 16.24 ms per query on standard CPU infrastructure.

---------- END OF INVENTION DISCLOSURE FORMAT ----------

References:

A. Patent Prior Art (cited in Section 3):

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- CN109063765B — Image Classification Method Based on Gated Neural Network Information Fusion. https://patents.google.com/patent/CN109063765B/en
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- US12511322 — Large Language Model-Assisted Entity Name Resolution. https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/12511322

B. Technical / Non-Patent Literature (underlying methodologies):

- Reimers, N. and Gurevych, I., 'Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,' Proceedings of EMNLP-IJCNLP, 2019. (Underlying sentence-transformer dense embedding foundation).

- Chen, J. et al., 'BGE M3-Embedding: Multi-Lingual, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation,' 2024. (Underlying BGE-M3 vector architecture).
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- Cui, Y. et al., 'Class-Balanced Loss Based on Effective Number of Samples,' IEEE CVPR, 2019. (Class-balancing weight formulation applied to imbalanced CPC subclasses).
- Johnson, J., Douze, M., and Jégou, H., 'Billion-scale Similarity Search with GPUs,' IEEE Transactions on Big Data, 2019. (Meta FAISS dense vector search index implementation).
- Robertson, S. and Zaragoza, H., 'The Probabilistic Relevance Framework: BM25 and Beyond,' Foundations and Trends in Information Retrieval, 2009. (Sparse lexical baseline benchmark).
- Lewis, P. et al., 'Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,' NeurIPS, 2020. (Retrieval-Augmented Generation paradigm for grounded question answering).
- WIPO, 'Guide to the International Patent Classification and Cooperative Patent Classification (CPC),' World Intellectual Property Organization & EPO, 2024. (Taxonomy for 15 evaluated CPC subclasses).